8 Projects (optional, for fun)

QUICK

Project 6. A soil moisture sensor that tells you when a bed is dry

Quick version

No time to read it all? This is the short version. The full guide is below.

- A small ESP32 board with a soil sensor sends the moisture to Home Assistant every minute. Your phone gets a message when the bed is dry.
- Medium, about 2 hours, about $15. Do Project 3 first.
- You need an ESP32 board, a capacitive soil sensor, a USB cable that carries data, and 2.4 GHz Wi-Fi.
- Start ESPHome: `docker compose -f docker/extras/esphome.yml up -d`, then open http://localhost:6052.
- Wire the sensor to 3V3, GND and GPIO34, with the board unplugged. Only the long thin part goes in the soil.
- Measure your own dry and wet numbers (Part E). The ESP32 must never get wet.
- STOP SIGN: the sensor and the ESP32 run on 3.3 V from USB. Nothing here touches wall power.